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import pandas as pd
import numpy as np
from matplotlib import pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix,
accuracy_score
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import StandardScaler

from imblearn.over_sampling import SMOTE

# Load data
water = pd.read_csv("water_potability.csv")

# Fix missing values
water.fillna(water.mean(numeric_only=True), inplace=
True)

# Features & labels
f = water.drop('Potability', axis=1)
l = water['Potability']

A = np.array(f)
B = np.array(l)

# Handle imbalance
sm = SMOTE(random_state=42)
A_res, B_res = sm.fit_resample(A, B)

# Scaling
scaler = StandardScaler()
A_scaled = scaler.fit_transform(A_res)

# Split
xtrain, xtest, ytrain, ytest = train_test_split(
A_scaled, B_res, test_size=0.3, random_state=42)
```

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39
40 # Models
41 model1 = LogisticRegression(max_iter=500)
42 model2 = GaussianNB()
43 model3 = KNeighborsClassifier(n_neighbors=7)
44 model4 = RandomForestClassifier(n_estimators=200,
   random_state=42)
45
46 models = [model1, model2, model3, model4]
47 names = ["Logistic Regression", "Naive Bayes", "KNN"
   , "Random Forest"]
48
49 results = []
50
51 for model, name in zip(models, names):
52     model.fit(xtrain, ytrain)
53     pred = model.predict(xtest)
54
55     acc = accuracy_score(ytest, pred)
56     cm = confusion_matrix(ytest, pred)
57
58     print(f"\n{name}")
59     print("Confusion Matrix:\n", cm)
60     print("Accuracy:", acc)
61
62     results.append(acc)
63
64 # Visualization
65 df = pd.DataFrame({"Model": names, "Accuracy":
   results})
66
67 sns.barplot(x="Model", y="Accuracy", data=df,hue='
   Model')
68 plt.show()
69 print(df)
70
```